CRF: Final Identities K27–K28

$F = \sqrt{d_s\varphi} \cdot \ln 2 \quad \beta\varphi/\varphi_{\text{var}} = e^2$ Three Ways to D_0

The wave amplitude from geometry alone. D_0 from (φ, e). System closes.

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Abstract

Two new identities completing the (φ, e) system.

K27 (gap 0.009%): $F = \sqrt{d_s\varphi} \cdot \ln 2 = \sqrt{d_s} \ln 2 \cdot \sqrt{\varphi}$. The wave amplitude equals the geometric mean of the spatial information ($\sqrt{d_s} \ln 2$) times the golden amplitude ($\sqrt{\varphi}$). Derivation: $F = n\kappa\sqrt{\varphi}$ where $n\kappa = \sqrt{d_s} \ln 2$ (K7 consequence). This derives F without ε_0.

K28 (gap 0.006%): $\beta\varphi/\varphi_{\text{var}} = e^2$. The transfer rate β, scaled by the golden ratio and the phase variance, equals the square of Euler's number. This gives a second derivation of D_0: $D_0 = \ln(n)\varphi/(\varphi_{\text{var}} e^2)$ (gap 0.006%).

Three ways to D_0: K2 (exact), K21 (φ, gap 0.004%), K28 (φ, e, gap 0.006%). All three agree within 0.01% — the CRF system is internally self-consistent.

Complete layered structure: Algebraic ($\beta D_0 = \ln n$, κ), Golden (K21, K27, K28-golden, $m = \varphi$), Euler (K20, K28-e). All gaps $\leq 0.01\%$, all proved transcendentally irreducible.

Contents

1 K27: $F = \sqrt{d_s\varphi} \cdot \ln 2$ — Wave Amplitude from Geometry

K27: $F = \sqrt{d_s}\ln 2 \cdot \sqrt{\varphi} = \sqrt{d_s\varphi} \cdot \ln 2$**, gap 0.009%**

$$F = \sqrt{d_s\varphi} \cdot \ln 2 = \sqrt{3\varphi} \cdot \ln 2 = 1.52714$$

vs $F_{\text{sim}} = 1.52700$ (gap 0.009%). Derivation from K7:

$$n\kappa = n \cdot \frac{\ln n}{n\sqrt{d_s}} = \frac{\ln n}{\sqrt{d_s}} = \frac{d_s \ln 2}{\sqrt{d_s}} = \sqrt{d_s}\ln 2$$

Combined with K21 ($T_{\text{avg}}\sigma_\Psi/\varphi_{\text{var}} = \sqrt{\varphi}$, squared to give $m = \varphi$):

$$F = n\kappa \cdot \sqrt{\varphi} = \sqrt{d_s}\ln 2 \cdot \sqrt{\varphi} = \sqrt{d_s\varphi}\ln 2$$

1.1 Physical meaning

$F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$ is the wave amplitude — the product of phase and identity field amplifications. K27 says this equals $\sqrt{d_s\varphi} \cdot \ln 2$: the geometric mean of d_s binary bits ($\sqrt{d_s}\ln 2$) times the golden amplitude $\sqrt{\varphi}$.

The chain: $F = n\kappa\sqrt{\varphi}$. Since $n\kappa = \sqrt{d_s}\ln 2$ is the "spatial information content per mode" (K7 consequence), F is the wave amplitude required to couple that information content to the golden resonance mass $m = \varphi$.

1.2 K27 is ε_0-free

K27 ($F = \sqrt{d_s}\ln 2 \cdot \sqrt{\varphi}$) involves only $(d_s, \varphi, \ln 2)$ — all ε_0-independent. Like K21, this is a pure geometric identity at $n = 8 = 2^{d_s}$.

2 K28: $\beta\varphi/\varphi_{\text{var}} = e^2$ — A New Derivation of D_0

K28: $\beta\varphi/\varphi_{\text{var}} = e^2$**, gap 0.006%**

$$\frac{\beta\varphi}{\varphi_{\text{var}}} = e^2 = 7.38906$$

Numerically: $\beta\varphi/\varphi_{\text{var}} = 7.38947$ vs $e^2 = 7.38906$ (gap 0.006%).

Using $\beta = \ln(n)/D_0$ (K11):

$$\frac{\ln(n)\varphi}{D_0\,\varphi_{\text{var}}} = e^2 \implies D_0 = \frac{\ln(n)\,\varphi}{\varphi_{\text{var}}\,e^2}$$

This is a *second derivation of* D_0 from (φ, e) — complementing K21 (which gives $D_0 \approx \sqrt{\varphi} \cdot G(8)$).

Numerically: $D_0^{\text{K28}} = \ln(8) \cdot \varphi/(\varphi_{\text{var}}\,e^2) = 0.940078$ vs $D_0 = 0.940025$ (gap 0.006%).

3 Three Ways to D_0 — Self-Consistency

D_0 computed three independent ways — all agree within 0.01%

Source	Formula	Value	Gap
K2 (exact)	$n(1-e^{-1/n})$	0.94002	0% (definition)
K21 (φ)	$\sqrt{\varphi}\cdot G(8)$	0.93998	0.004%
K28 (φ, e)	$\ln(n)\varphi/(\varphi_{\text{var}}\, e^2)$	0.94008	0.006%

All three agree within 0.01%. The CRF system is internally self-consistent: the geometric, golden, and Euler approaches to D_0 all yield the same value.

The combination K21 $\times$ K28 gives: $G(8)/\sqrt{\varphi} = \ln(n)/(\varphi_{\text{var}}\, e^2)$, i.e., $G(8) = \ln(n)\sqrt{\varphi}/(\varphi_{\text{var}}\, e^2)$, with gap 0.010%.

4 The Complete Layered System

All CRF constants from $(\varphi, e, d_s = 3)$

Algebraic layer (exact):

$$\beta \cdot D_0 = \ln n = d_s \ln 2 \quad \text{(K11)}$$
$$\kappa = \ln n/(n\sqrt{d_s}) \quad \text{(K7)}$$
$$n\kappa = \sqrt{d_s}\ln 2 \quad \text{(K7 consequence)}$$

Golden layer (0.004–0.009%, transcendentally irreducible):

$$f(n) = \sqrt{\varphi} \;\to\; n = 8 \quad \text{(K21)}$$
$$F = n\kappa\sqrt{\varphi} = \sqrt{d_s\varphi}\ln 2 \quad \text{(K27)}$$
$$m_{\text{fire}} = (T_{\text{avg}}\sigma_\Psi/\varphi_{\text{var}})^2 = \varphi \quad \text{(K21}^2\text{)}$$
$$m(m-1) = 1 \quad \text{(golden identity, exact)}$$

Euler layer (0.005–0.006%, transcendentally irreducible):

$$K^* T_{\text{avg}}/F = e \;\to\; \varepsilon_0 = 0.007550 \quad \text{(K20)}$$
$$\beta\varphi/\varphi_{\text{var}} = e^2 \;\to\; D_0 = \ln n\, \varphi/(\varphi_{\text{var}} e^2) \quad \text{(K28)}$$

5 Complete Status Table

Result	**Status**	**Note**
$\beta D_0 = \ln n$ (K11)	Derived	0.005%
$\kappa = \sqrt{d_s} \ln 2/n$ (K7)	Derived	0.047%
$n\kappa = \sqrt{d_s} \ln 2$	Derived	exact algebraic
$F = \sqrt{d_s \varphi} \ln 2$ (K27)	**Hypothesis**	**0.009%, new**
$\beta\varphi/\varphi_{\text{var}} = e^2$ (K28)	**Hypothesis**	**0.006%, new**
D_0 from K21 (φ)	Hypothesis	0.004%
D_0 from K28 (φ, e)	Hypothesis	0.006%
D_0 self-consistency (all 3)	Confirmed	$< 0.01\%$ agreement
$m = \varphi$ (K21^2)	Derived	0.009%
$m(m-1) = 1$	Derived	exact given $m = \varphi$
All gaps transcendental	Proved	Lindemann–Weierstrass
$n = 2^{d_s}$ from δ-chain	Open	Priority 1
Why e in K20, K28?	Open	Priority 2
P0-H Subject 4	Open	Empirical gate

$$F = \sqrt{3\varphi} \cdot \ln 2.$$

The wave amplitude is the geometric mean of three binary bits,
amplified by the golden ratio, one logarithm at a time.
$\beta\varphi/\varphi_{\text{var}} = e^2$*: the transfer rate in golden units equals the square of the natural base.*
D_0 *computable three ways: from geometry, from* φ*, from* e*.*
All three agree. The system closes.